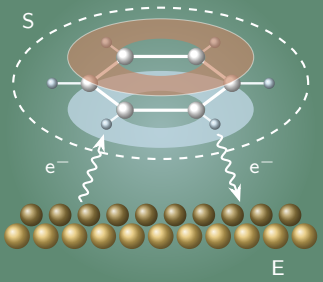
Open Quantum Systems

When molecules exchange electrons with their environment

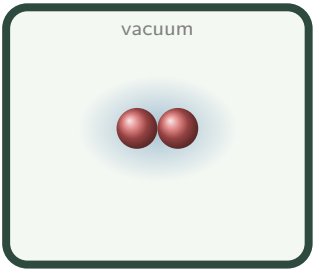


Gabriele Fabbro

Department of Chemistry and Biomedical Science
Norwegian University of Science and Technology (NTNU)

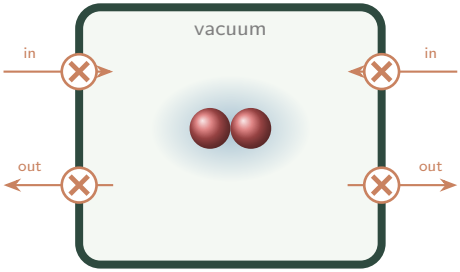
NTNU

How we usually model a molecule: a closed system



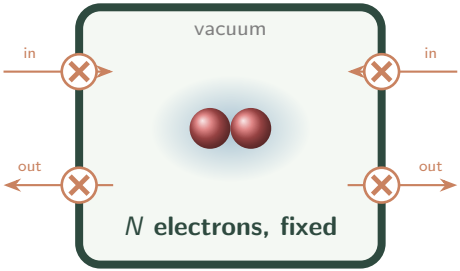
- The molecule is alone, in vacuum.

How we usually model a molecule: a closed system



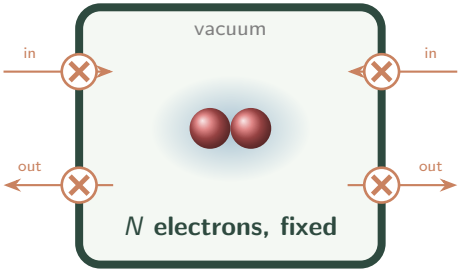
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How we usually model a molecule: a closed system

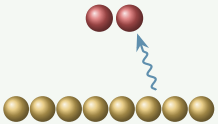


- The molecule is alone, in vacuum.
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$\hat{H}\,\Psi = E\,\Psi$, with N fixed.
The starting point of most quantum chemistry.

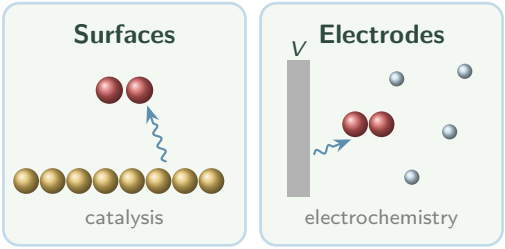
Reality: molecules are rarely alone

Surfaces

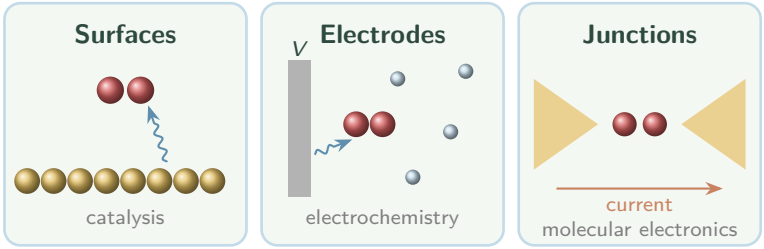


catalysis

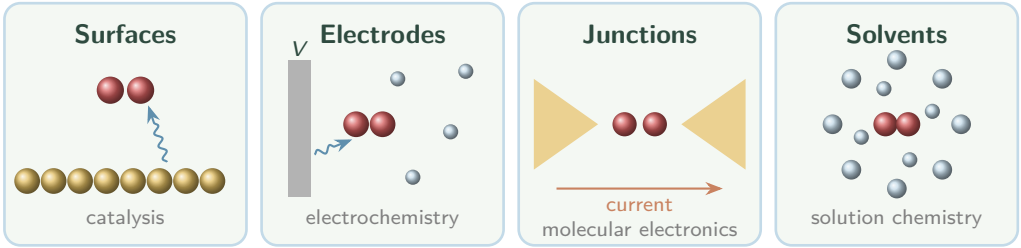
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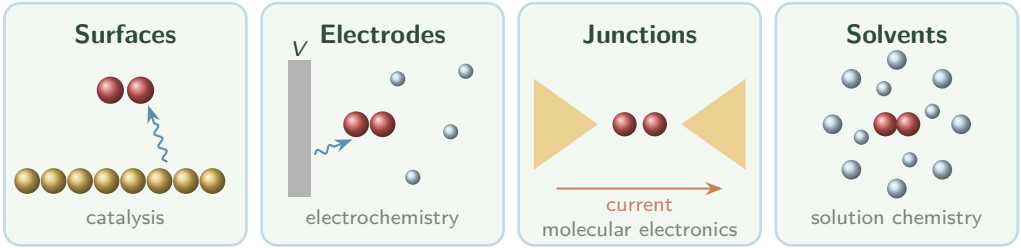
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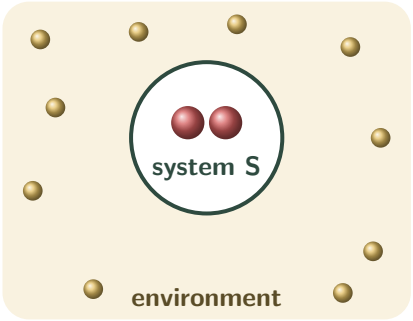


Reality: molecules are rarely alone



They exchange **electrons** and **energy** with their surroundings: **open quantum systems**.

System and environment

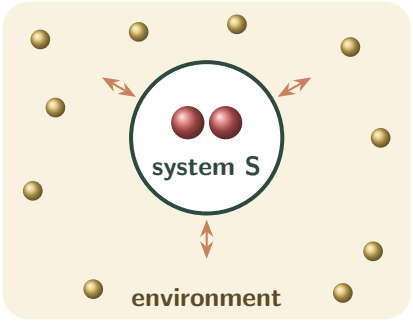


The Hamiltonian of everything has three parts:

$$\hat{H} = \underbrace{\hat{H}_{\mathrm{S}}}_{\text{molecule}} + \underbrace{\hat{H}_{\mathrm{E}}}_{\text{environment}} + \underbrace{\hat{H}_{\mathrm{SE}}}_{\text{coupling}}$$

H.-P. Breuer, F. Petruccione, *The Theory of Open Quantum Systems* (Oxford University Press, 2002).

System and environment



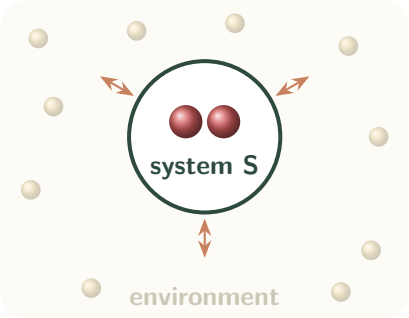
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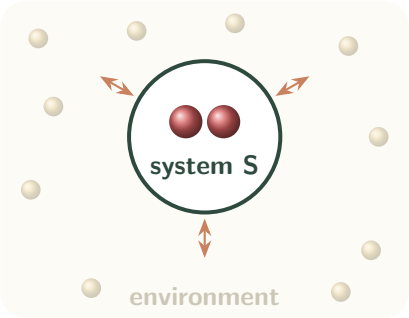
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The Hamiltonian of everything has three parts:

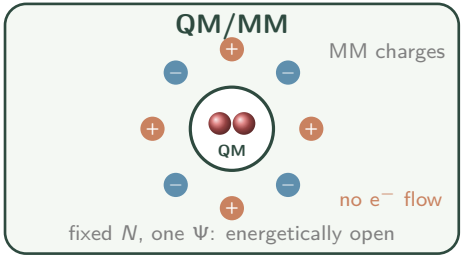
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Key question:
does the coupling only polarize S,
or does it also move electrons?

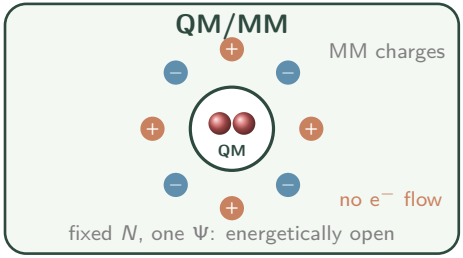
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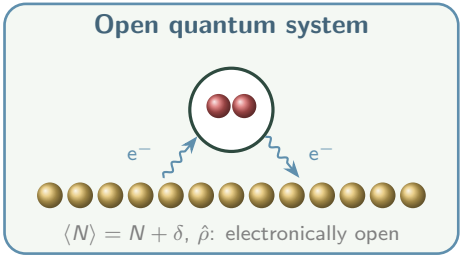
Open quantum systems are not QM/MM



H. M. Senn, W. Thiel, *Angew. Chem. Int. Ed.* **48**, 1198 (2009); B. S. Sannes, J. Pedersen, I.-M. Høyvik, *J. Chem. Phys.* **162**, 094108 (2025); J. Pedersen *et al.*, *J. Phys. Chem. A* **129**, 4288 (2025).

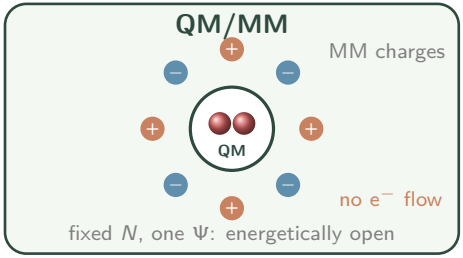
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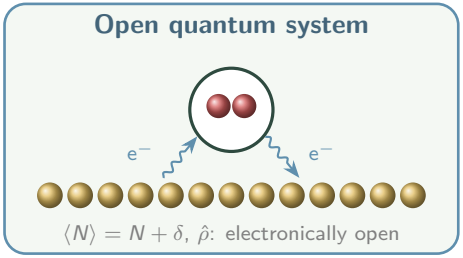




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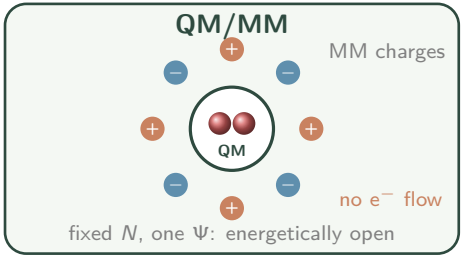
The coupling has two parts:

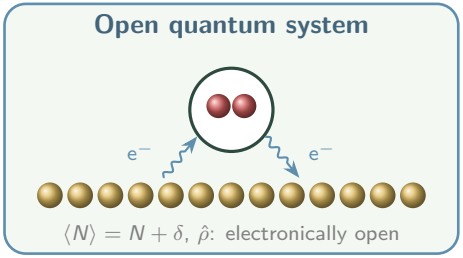
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Particle-conserving: electrostatics, polarization. Particle-breaking: electron exchange.

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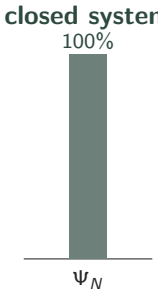
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Complementary: QM/MM for electrostatics, open-system theory for electron exchange.

H. M. Senn, W. Thiel, *Angew. Chem. Int. Ed.* **48**, 1198 (2009); B. S. Sannes, J. Pedersen, I.-M. Høyvik, *J. Chem. Phys.* **162**, 094108 (2025); J. Pedersen *et al.*, *J. Phys. Chem. A* **129**, 4288 (2025).

The density operator

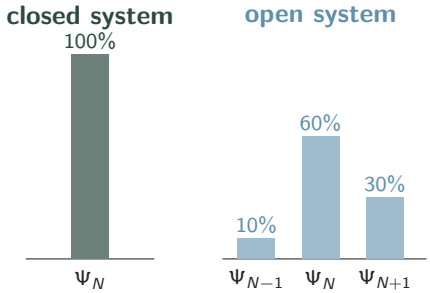


possible states and their probabilities

- Closed: one state, known exactly.

J. von Neumann, *Gött. Nachr.* 245 (1927); U. Fano, *Rev. Mod. Phys.* **29**, 74 (1957); H.-P. Breuer, F. Petruccione, *The Theory of Open Quantum Systems* (Oxford University Press, 2002).

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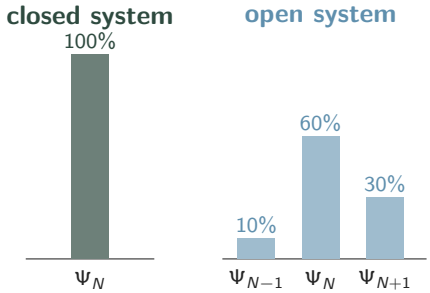


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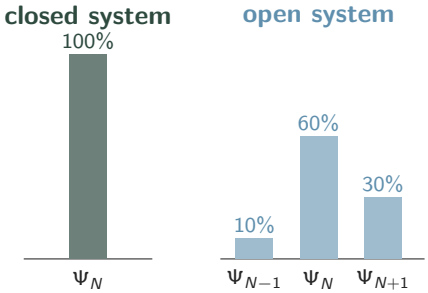
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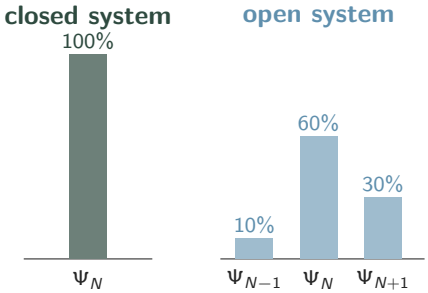
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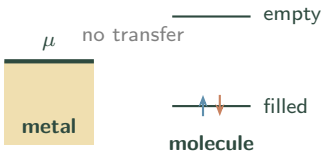
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Liouville–von Neumann equation:
$i\hbar \, \frac{d\hat{\rho}}{dt} = [\hat{H}, \hat{\rho}]$.

J. von Neumann, *Gött. Nachr.* 245 (1927); U. Fano, *Rev. Mod. Phys.* **29**, 74 (1957); H.-P. Breuer, F. Petruccione, *The Theory of Open Quantum Systems* (Oxford University Press, 2002).

Phenomenon 1: level alignment and electron transport

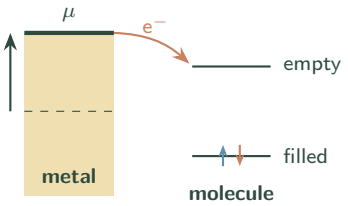
One electrode: level alignment



- The metal is filled up to its Fermi level μ.

Phenomenon 1: level alignment and electron transport

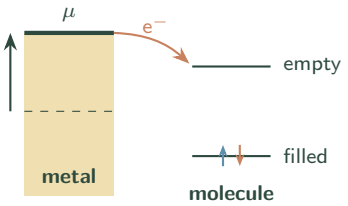
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- μ above the empty level: electrons enter.
- It sets injection barriers and redox potentials.

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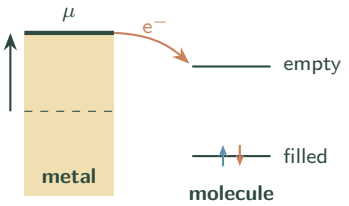
Two electrodes: electron transport



- The same μ on both sides: no current.

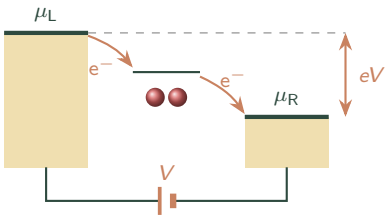
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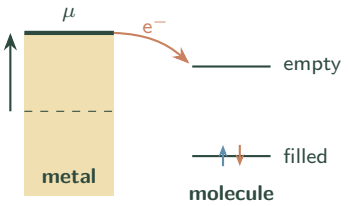
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- A voltage V gives $\mu_L - \mu_R = eV$.
- Electrons hop in on one side, out on the other.

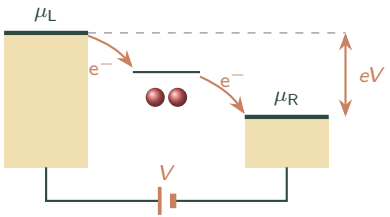
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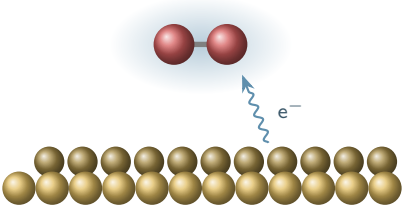
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With an open model: each electrode is a reservoir at its own μ.
One reservoir gives the charging curve $\langle N \rangle(\mu)$. Two reservoirs give the current.

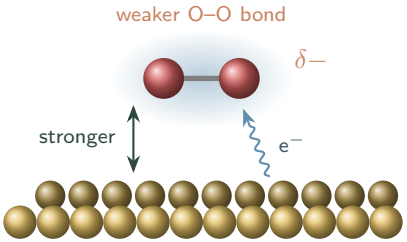
Phenomenon 2: partial charge



What happens

- A molecule adsorbs on a metal surface.
- Its levels mix with those of the metal.

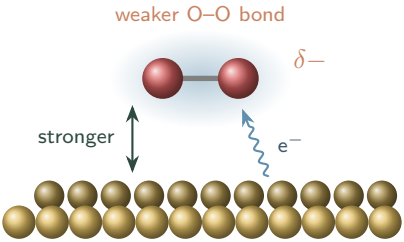
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- The bond to the surface gets stronger.
- The bond inside the molecule gets weaker.

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- On average it carries a fraction of an electron.
- The bond to the surface gets stronger.
- The bond inside the molecule gets weaker.

With an open model
The charge is a result: $\langle N \rangle = \mathrm{Tr}\big(\hat{\rho}\,\hat{N}\big)$.
It follows from μ and from the coupling.

Phenomenon 3: activation of small molecules

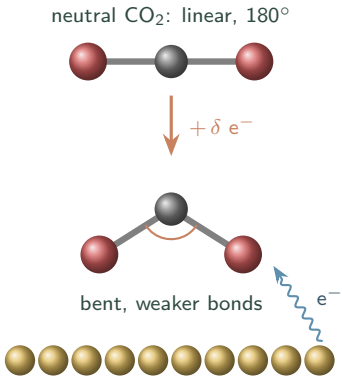
neutral CO_2: linear, 180°



What happens

- Neutral CO_2 is linear and very stable.

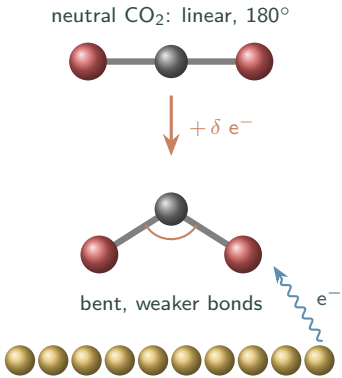
Phenomenon 3: activation of small molecules



What happens

- Neutral CO_2 is linear and very stable.
- Extra charge goes into an antibonding orbital.
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- It is the first step of CO_2 reduction.

Phenomenon 3: activation of small molecules



What happens

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With an open model
A molecule at a fractional charge.
We follow geometry and bonds versus charge.

The OpenQuantum project



Ida-Marie Høyvik
ERC Consolidator Grant

Molecules as electronically open quantum systems

- Molecules share or exchange electrons.
- This changes their properties.
- Electron currents can pass through them.
- Goal: models where N is **not** fixed.

European Research Council
Established by the European Commission

ERC Consolidator Grant
OpenQuantum, grant agreement No. 101170817

CORDIS, project 101170817, *Molecules as electronically open quantum systems* (2025–2030).

Take-home messages

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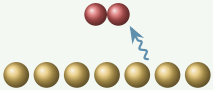
4. We want more than a fractional charge: the state of the open molecule.
   OpenQuantum builds models where N is not fixed.

Master projects in the quantum chemistry group

Supervisors: **Sarai D. Folkestad**, **Ida-Marie Høyvik** and **Henrik Koch**.



Polaritonic chemistry
Molecules coupled to light.



Charge transfer & bonding
Charge at interfaces.



Photochemistry
Molecules driven by light.



Large systems
Accuracy where it matters.



Software (eT)
From equations to code.

Thank you! Questions?

Why not just a fractional N, or Mulliken charges?

Fractional N as input

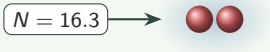


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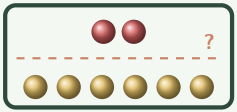
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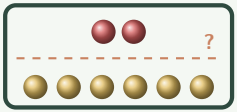
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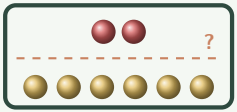
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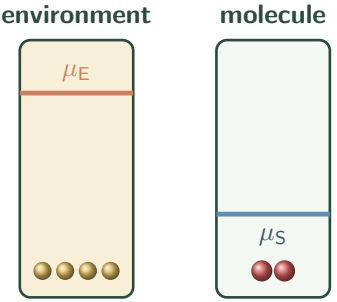


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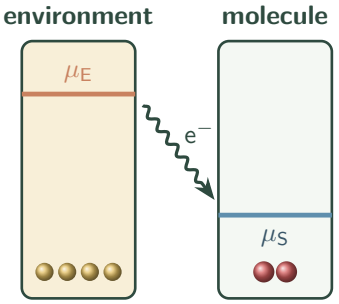
Why the chemical potential matters



- $\mu = \partial E/\partial N$: the energy per added electron.

R. G. Parr, R. A. Donnelly, M. Levy, W. E. Palke, *J. Chem. Phys.* **68**, 3801 (1978); R. G. Parr, R. G. Pearson, *J. Am. Chem. Soc.* **105**, 7512 (1983). μ is minus the electronegativity, $\mu = -\chi$.

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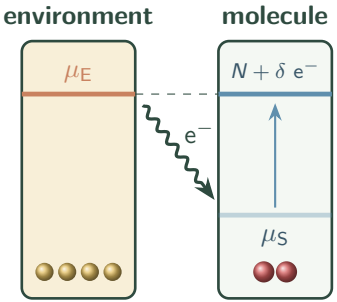


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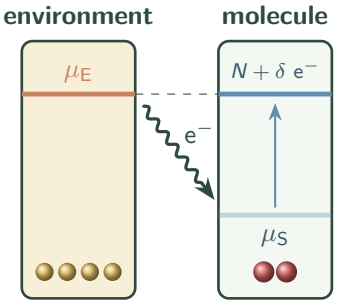


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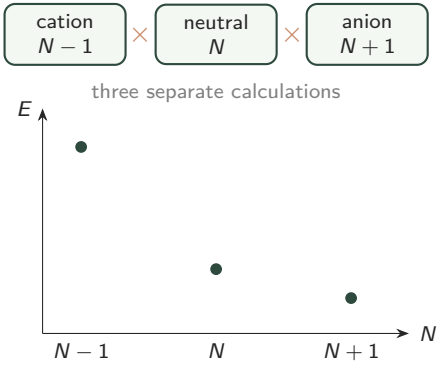
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The environment acts as a reservoir at μ_E.
We need μ of the molecule: how is it computed?

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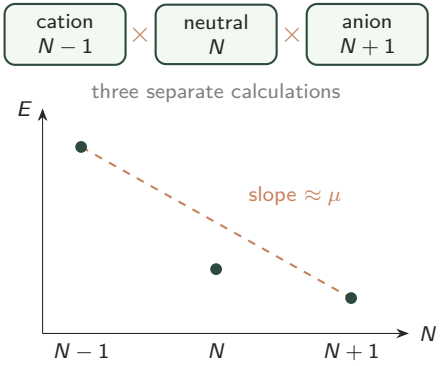
The closed picture: how μ has been computed



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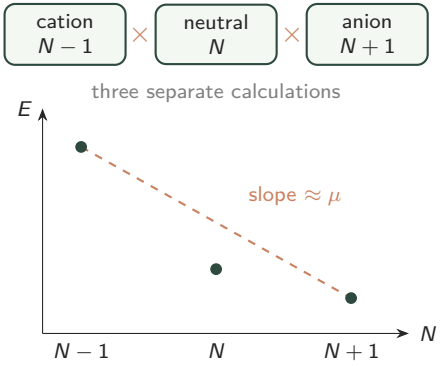
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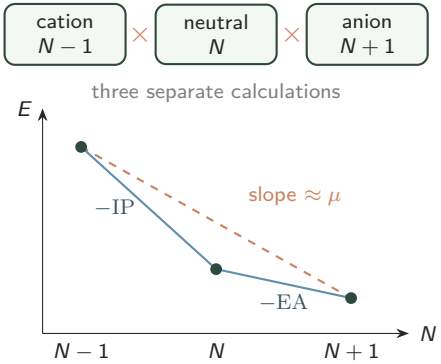
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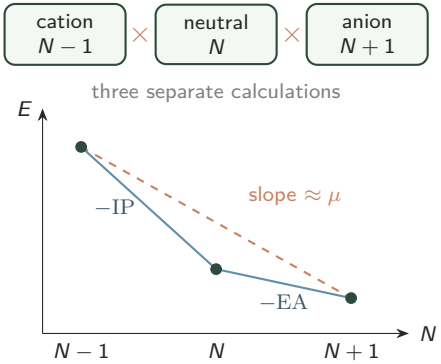


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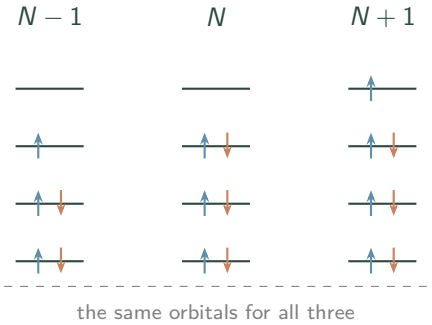
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We want one calculation, a fractional N and μ on each side: **GCUHF**.

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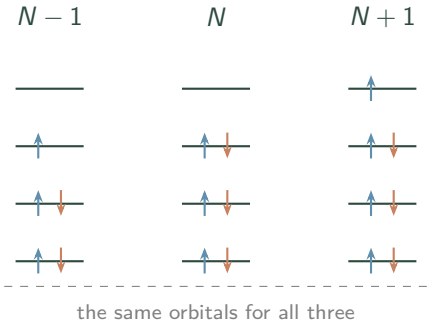
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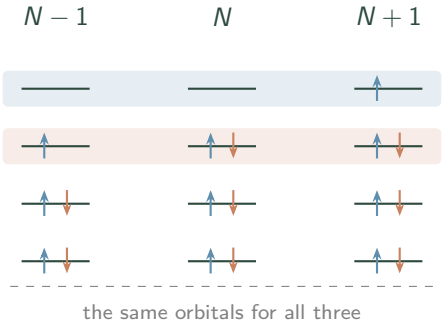
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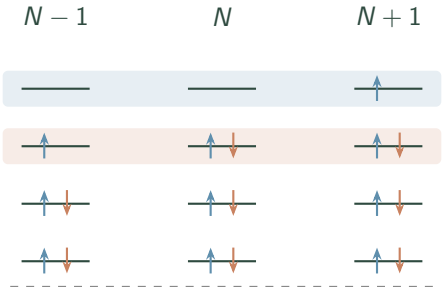
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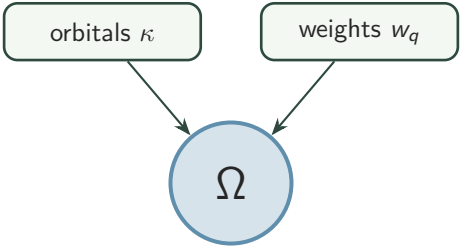


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$\hat{\rho} = \sum_q w_q \, |\Phi_q\rangle\langle\Phi_q|, \;\; q \in \{N-1, N, N+1\}.$
Unknowns: the orbitals and the weights w_q.

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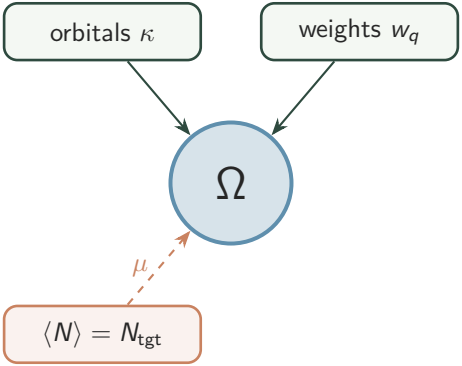
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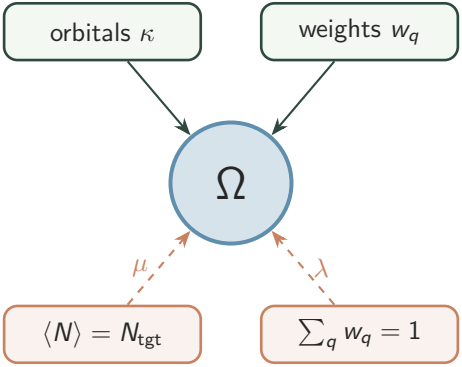
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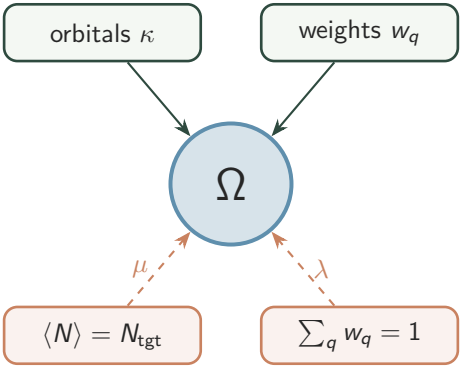
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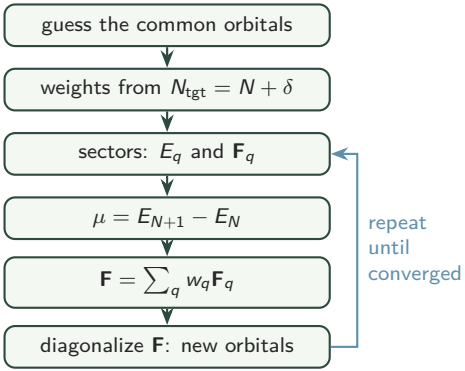


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- As in thermodynamics: a **grand potential**.

$$\Omega[\kappa, w, \mu, \lambda] = \sum_q w_q E_q[\kappa] - \mu\Big(\sum_q w_q N_q - N_{\text{tgt}}\Big) - \lambda\Big(\sum_q w_q - 1\Big)$$

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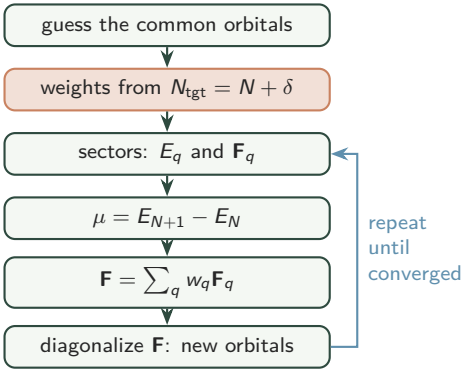
Minimizing Ω: orbitals, weights and μ



- Active sectors: N and $N + 1$.

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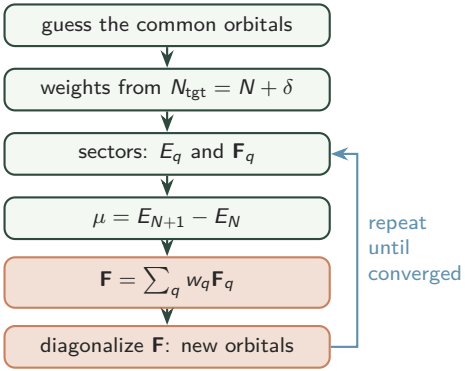
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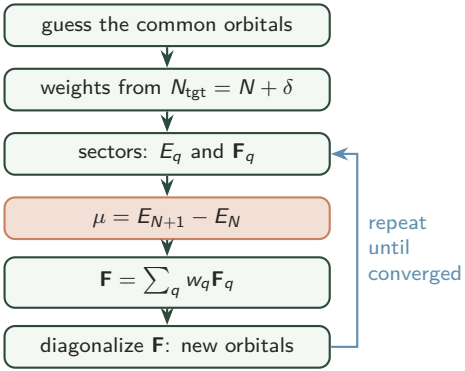
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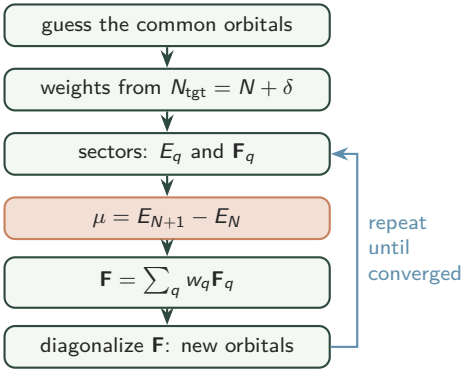
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- μ: $\partial\Omega/\partial w_q = 0$ for both active sectors.

$$E_{N+1} - \mu\,(N+1) - \lambda = 0$$
$$E_N - \mu\,N - \lambda = 0$$

Subtract: λ cancels, and $\boxed{\mu = E_{N+1} - E_N}$.

G. Fabbro, I.-M. Høyvik, *Grand-canonical ensemble mean-field theory* (in preparation); P. Pulay, *Chem. Phys. Lett.* **73**, 393 (1980) (DIIS).
Below N the active sectors are $N-1$ and N, and $\mu = E_N - E_{N-1}$.

Minimizing Ω: orbitals, weights and μ



- Active sectors: N and $N+1$.
- Weights: $w_N = 1 - \delta$, $w_{N+1} = \delta$.
- Orbitals: one Fock matrix, as in Hartree–Fock.
- μ: $\partial\Omega/\partial w_q = 0$ for both active sectors.

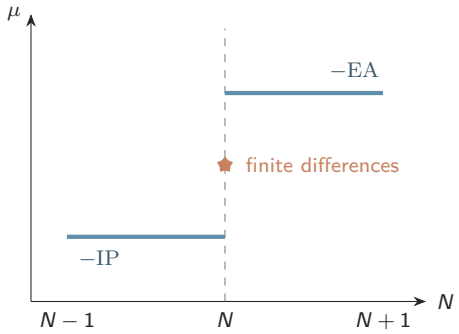
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Subtract: λ cancels, and $\boxed{\mu = E_{N+1} - E_N}$.

At convergence, $\mu = dE/dN = \varepsilon_{\text{frontier}}$.

G. Fabbro, I.-M. Høyvik, *Grand-canonical ensemble mean-field theory* (in preparation); P. Pulay, *Chem. Phys. Lett.* **73**, 393 (1980) (DIIS).
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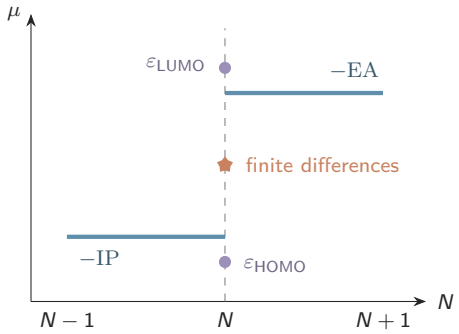
μ computed in different ways: what is equal, what differs



Method	What it gives
Finite differences	midpoint of the two branches
ΔSCF, two calculations	average of μ on one branch

Schematic; numbers for K and O_2 are in the backup slides. FON-UHF equality: active sectors differing by one spin-orbital.

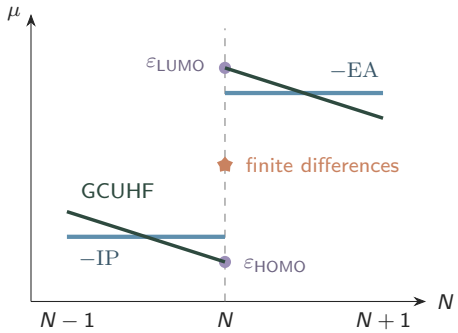
μ computed in different ways: what is equal, what differs



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Schematic; numbers for K and O_2 are in the backup slides. FON-UHF equality: active sectors differing by one spin-orbital.

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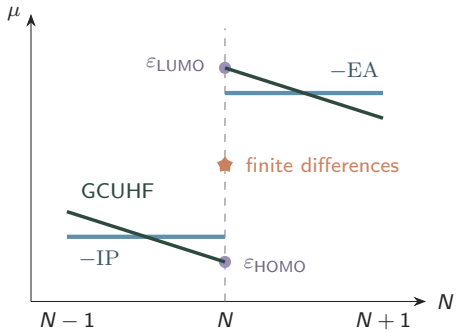


Method	What it gives
Finite differences	midpoint of the two branches
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Koopmans	μ at the two ends, at N
FON-UHF	the same μ as GCUHF
GCUHF	μ at every fractional N

GCUHF connects them, in one calculation.
Its ends are Koopmans, its average is ΔSCF.

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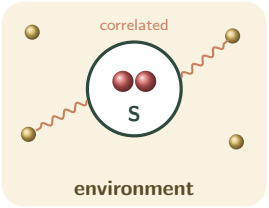
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Its ends are Koopmans, its average is ΔSCF.

Limits: Hartree–Fock level, so EA is poor.
μ drifts along a branch, and N is imposed.

Schematic; numbers for K and O_2 are in the backup slides. FON-UHF equality: active sectors differing by one spin-orbital.

Appendix: why tracing out the environment gives a mixture

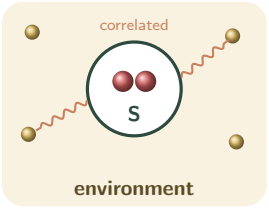


An electron shared by S and E (total $N + M$ fixed):

$$|\Psi_{\text{tot}}\rangle = c_1\,|N\rangle_{\text{S}}|M\rangle_{\text{E}} + c_2\,|N{+}1\rangle_{\text{S}}|M{-}1\rangle_{\text{E}}$$

H.-P. Breuer, F. Petruccione, *The Theory of Open Quantum Systems* (Oxford University Press, 2002); R. Matveeva *et al.*, *J. Phys. Chem. A* **127**, 1329 (2023), eqs. 15–17; J. Pedersen, B. S. Sannes, I.-M. Høyvik, *J. Chem. Theory Comput.* **21**, 9415 (2025).

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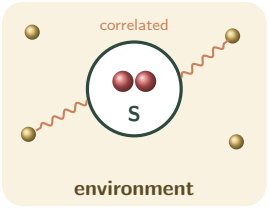
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We do not look at E: partial trace.

$$\hat{\rho}_{\text{S}} = |c_1|^2\,|N\rangle\langle N| + |c_2|^2\,|N{+}1\rangle\langle N{+}1| + \Big(c_1 c_2^*\,\langle M{-}1|M\rangle_{\text{E}}\,|N\rangle\langle N{+}1| + \text{h.c.}\Big)$$

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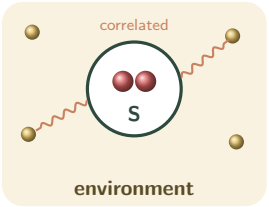
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Correlated S and E $\Rightarrow$ mixed $\hat{\rho}_{\text{S}}$.
The information is in the S–E correlations.
Purity: $\operatorname{Tr}\hat{\rho}_{\text{S}}^2 < 1$. Density-operator route.

Product state $|\psi\rangle_{\text{S}}|\chi\rangle_{\text{E}} \Rightarrow$ pure $\hat{\rho}_{\text{S}}$.
PBHF assumes $C_{IJ} \approx C_I^{\text{A}} C_J^{\text{B}}$.
S keeps one wavefunction with mixed N.

H.-P. Breuer, F. Petruccione, *The Theory of Open Quantum Systems* (Oxford University Press, 2002); R. Matveeva *et al.*, *J. Phys. Chem. A* **127**, 1329 (2023), eqs. 15–17; J. Pedersen, B. S. Sannes, I.-M. Høyvik, *J. Chem. Theory Comput.* **21**, 9415 (2025).

Appendix: two ways to open a molecule, pros and cons

Particle-breaking HF: one wavefunction

$$|\Psi\rangle = e^{\hat{\gamma}}|\Phi\rangle, \quad \hat{H} = \hat{H}_{\text{mol}} + \hat{H}_{\text{pb}}$$

- ✓ One normalized wavefunction.
- ✓ Mean-field cost; HF without coupling.
- ✓ Size-extensive; response via TDPBHF.
- × Coupling only via model parameters λ_p.
- × Pure state: no dissipation.
- × Lifetimes need ad hoc damping.
- × Odd charges need bath orbitals.

R. Matveeva *et al.*, *J. Phys. Chem. A* **127**, 1329 (2023); R. Paul *et al.*, *J. Phys. Chem. A* **128**, 1533 (2024); J. Pedersen *et al.*, *J. Phys. Chem. A* **129**, 4288 (2025); B. S. Sannes, J. Pedersen, I.-M. Høyvik, *J. Chem. Phys.* **162**, 094108 (2025); J. Pedersen, B. S. Sannes, I.-M. Høyvik, *J. Chem. Theory Comput.* **21**, 9415 (2025).

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Reduced density operator: a mixture

$$\hat{\rho}_{\text{S}} = \text{Tr}_{\text{E}}\,\hat{\rho}, \quad \text{from } \hat{H} = \hat{H}_{\text{M}} + \hat{H}_{\text{E}} + \hat{H}_{\text{ME}}$$

- ✓ Derived from molecule + environment.
- ✓ Mixed states, dissipation, lifetimes.
- ✓ Generalizes the grand-canonical $\hat{\rho}$.
- ✓ Systematically improvable.
- × Exact form: as hard as the full system.
- × Approximations: weak coupling, Markov.
- × Many charge states: costly.
- × Some broadening still ad hoc.

R. Matveeva *et al.*, *J. Phys. Chem. A* **127**, 1329 (2023); R. Paul *et al.*, *J. Phys. Chem. A* **128**, 1533 (2024); J. Pedersen *et al.*, *J. Phys. Chem. A* **129**, 4288 (2025); B. S. Sannes, J. Pedersen, I.-M. Høyvik, *J. Chem. Phys.* **162**, 094108 (2025); J. Pedersen, B. S. Sannes, I.-M. Høyvik, *J. Chem. Theory Comput.* **21**, 9415 (2025).

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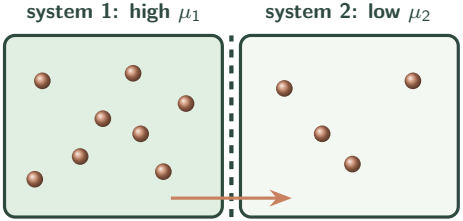
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GCUHF takes the density-operator route: a mixture of charge sectors, at mean-field cost.

R. Matveeva *et al.*, *J. Phys. Chem. A* **127**, 1329 (2023); R. Paul *et al.*, *J. Phys. Chem. A* **128**, 1533 (2024); J. Pedersen *et al.*, *J. Phys. Chem. A* **129**, 4288 (2025); B. S. Sannes, J. Pedersen, I.-M. Høyvik, *J. Chem. Phys.* **162**, 094108 (2025); J. Pedersen, B. S. Sannes, I.-M. Høyvik, *J. Chem. Theory Comput.* **21**, 9415 (2025).

Backup: what the chemical potential is



permeable wall, same temperature

The energy cost of adding one particle:

$$dU = T\,dS - p\,dV + \mu\,dN, \qquad \mu = \left(\frac{\partial U}{\partial N}\right)_{S,V}$$

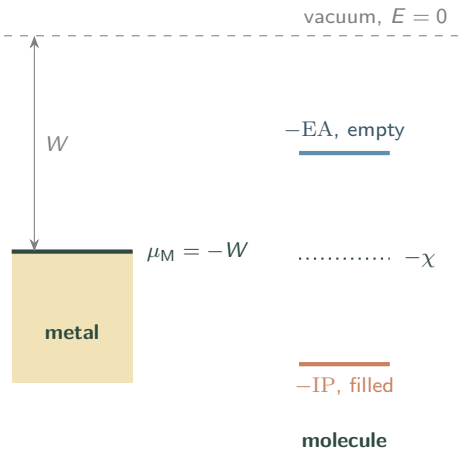
Two systems exchange particles, $dN_2 = -dN_1$:

$$dS_{\text{tot}} = \frac{\mu_2 - \mu_1}{T}\,dN_1 \;\geq\; 0$$

If $\mu_1 > \mu_2$, system 1 loses particles.
Equilibrium: $\mu_1 = \mu_2$.
μ is for particles what T is for heat.

J. W. Gibbs, *Elementary Principles in Statistical Mechanics* (Scribner, 1902).

Backup: electrons, the Fermi level and level alignment

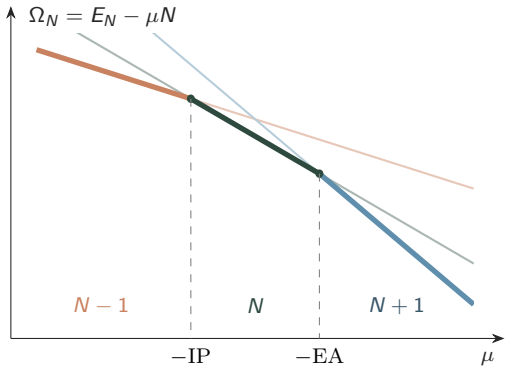


$\mathrm{IP} = E_{N-1} - E_N$, $\mathrm{EA} = E_N - E_{N+1}$, W: work function, $\chi = (\mathrm{IP} + \mathrm{EA})/2$.

- Molecule, fixed nuclei: $\mu = (\partial E/\partial N)_v$.
- Metal: μ_M is the Fermi level, $\mu_\mathrm{M} = -W$.
- $-\mathrm{EA} < \mu_\mathrm{M}$: electrons go to the molecule.
- $-\mathrm{IP} > \mu_\mathrm{M}$: electrons go to the metal.
- In between: the molecule stays neutral.

An electrode potential U shifts the Fermi level.
$\mu_\mathrm{M} \to \mu_\mathrm{M} - eU$: this is electrochemistry.

Backup: the grand-canonical ensemble at zero temperature



A molecule in contact with a reservoir at μ:

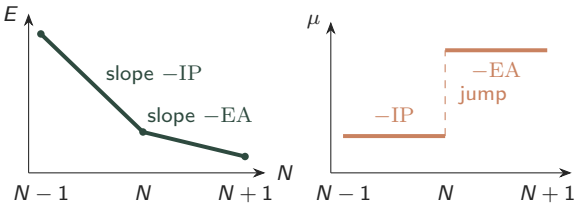
$$\hat{\rho} = \frac{e^{-\beta(\hat{H}-\mu\hat{N})}}{\Xi}, \qquad \langle \hat{N} \rangle = \frac{1}{\beta}\frac{\partial \ln \Xi}{\partial \mu}$$

- $T \to 0$: the sector with the lowest Ω_N wins.
- The lines have slope $-N$.
- The thick line is the lower envelope.

N electrons if and only if $-\mathrm{IP} < \mu < -\mathrm{EA}$.
The electron number changes in jumps.

Here μ is an input, fixed by the reservoir, and $\langle N \rangle$ is the result.

Backup: the exact energy is piecewise linear



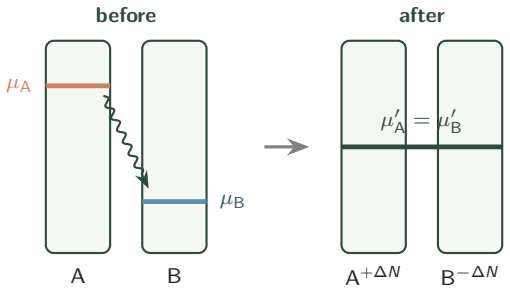
Fix the average $N + \delta$ instead of μ.
Minimize $\sum_N p_N E_N$ over the weights.
If E_N is convex, two neighbours win:

$$E(N+\delta) = (1-\delta)\, E_N + \delta\, E_{N+1}$$

$\mu = -\mathrm{IP}$ below N.
$\mu = -\mathrm{EA}$ above N.
At N: a jump of $\mathrm{IP} - \mathrm{EA}$.

J. P. Perdew, R. G. Parr, M. Levy, J. L. Balduz, *Phys. Rev. Lett.* **49**, 1691 (1982).

Backup: electronegativity, hardness and equalization



A smooth model through $N_0 - 1$, N_0, $N_0 + 1$:

$$E(N_0 + \Delta N) \approx E(N_0) + \mu\,\Delta N + \tfrac{1}{2}\eta\,\Delta N^2$$

$$\mu \approx -\frac{\mathrm{IP} + \mathrm{EA}}{2} = -\chi, \qquad \eta \approx \mathrm{IP} - \mathrm{EA}$$

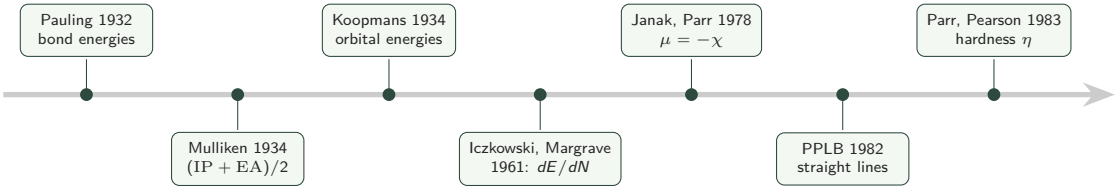
A and B in contact, minimize the total energy:

$$\Delta N = \frac{\mu_A - \mu_B}{\eta_A + \eta_B}$$

The transferred charge is fractional.

R. S. Mulliken, *J. Chem. Phys.* **2**, 782 (1934); R. T. Sanderson, *Science* **114**, 670 (1951); R. G. Parr *et al.*, *J. Chem. Phys.* **68**, 3801 (1978); R. G. Parr, R. G. Pearson, *J. Am. Chem. Soc.* **105**, 7512 (1983).

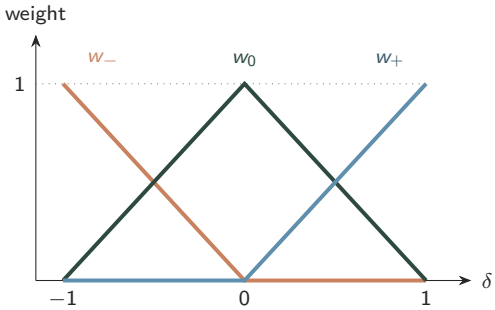
Backup: how μ has been computed, a short history



Method	**What it gives**	**Fractional N**	**Main limit**
Finite differences	$-(\mathrm{IP}+\mathrm{EA})/2$	no	three calculations, only the midpoint
Koopmans	$(\varepsilon_{\mathrm{HOMO}}+\varepsilon_{\mathrm{LUMO}})/2$	no	frozen orbitals, poor EA
Fractional occupations	slope $\partial E/\partial n_i=\varepsilon_i$	yes	curved $E(N)$ with approximate functionals
Grand-canonical DFT	N at a fixed μ	yes	errors of the functional
GCUHF	μ on each branch	yes	orbitals shared by the sectors

J. F. Janak, *Phys. Rev. B* **18**, 7165 (1978); R. Sundararaman, W. A. Goddard, T. A. Arias, *J. Chem. Phys.* **146**, 114104 (2017).

Backup: GCUHF, the weights are fixed by the constraints



$N_{\mathrm{tgt}} = N + \delta$

Two constraints, three weights:

$$w_- + w_0 + w_+ = 1, \qquad w_+ - w_- = \delta$$

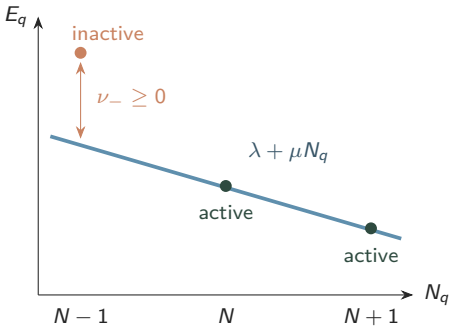
One free parameter, $s = w_- \geq 0$ (for $\delta \geq 0$):

$$E_{\mathrm{ens}}(s) = (1-\delta)E_0 + \delta E_+ + s\left(E_- + E_+ - 2E_0\right)$$

Convexity: $E_- + E_+ - 2E_0 \geq 0$, so $s = 0$.
Only two neighbouring sectors are active.
The only variational freedom is in the orbitals.

The weight problem is a linear program: its minimum lies at a vertex. Convexity is the condition $\mathrm{IP} \geq \mathrm{EA}$ on the common orbitals.

Backup: GCUHF, where μ comes from



μ is the slope of the line through the active sectors

Stationarity in the weights, with $w_q \geq 0$:

$$\frac{\partial \Omega}{\partial w_q} = E_q - \mu N_q - \lambda = \nu_q, \quad \nu_q \geq 0, \quad \nu_q w_q = 0$$

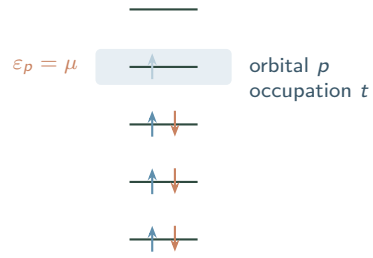
Active sectors N and $N+1$ have $\nu_q = 0$:

$$E_+ - \mu(N+1) - \lambda = 0$$
$$E_0 - \mu N - \lambda = 0$$

Subtract: $\mu = E_+ - E_0$, and $\lambda = E_0 - \mu N$.
Inactive sector: $\nu_- = E_- + E_+ - 2E_0 \geq 0$.

Karush–Kuhn–Tucker conditions. Below N the active sectors are $N-1$ and N, and $\mu = E_0 - E_-$.

Backup: GCUHF, μ is an orbital energy



The UHF energy is quadratic in the density:

$$E[\mathbf{P}_{q_1}+t\mathbf{D}] = E_{q_1}+t\,[\mathbf{F}_{q_1}]_{pp}+\tfrac{t^2}{2}\big(\underbrace{\mathrm{Tr}[\mathbf{DJ}(\mathbf{D})]}_{(pp|pp)}-\underbrace{\mathrm{Tr}[\mathbf{DK}(\mathbf{D})]}_{(pp|pp)}\big)$$

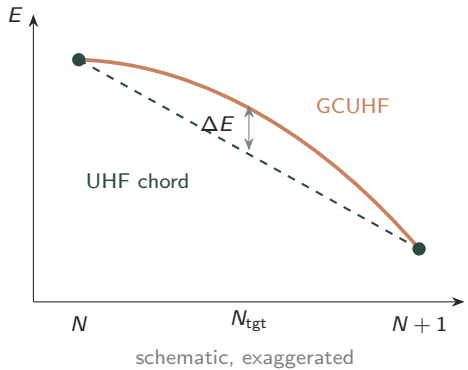
The self-repulsion of p cancels: the energy is linear in t.

$$\mu = E_{q_2} - E_{q_1} = [\mathbf{F}_{q_1}]_{pp} = [\mathbf{F}_{\mathrm{ens}}]_{pp} = \varepsilon_p$$

μ is the energy of the fractionally occupied orbital.
At fixed orbitals: the same energy as FON-UHF.

$\mathbf{F}_{\mathrm{ens}} = \mathbf{F}_{q_1} + t\,\mathbf{G}$ with $\mathbf{G} = \mathbf{J}(\mathbf{D}) - \mathbf{K}(\mathbf{D})$ and $G_{rp} = (rp|pp) - (rp|pp) = 0$. The two active sectors must differ by one spin-orbital.

Backup: GCUHF, μ is the slope and $E(N)$ is concave



On a branch, $E_{\text{ens}} = (1-t)E_{q_1}(\kappa) + t\,E_{q_2}(\kappa)$.
The orbitals are stationary, so:

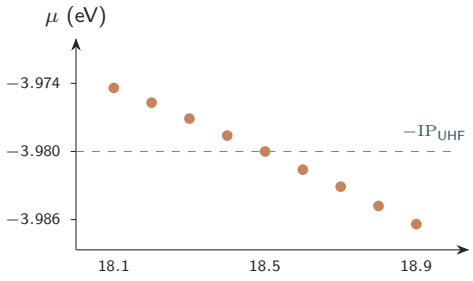
$$\frac{dE_{\text{GCUHF}}}{dt} = \underbrace{\frac{\partial E_{\text{ens}}}{\partial \kappa}}_{0}\frac{d\kappa}{dt} + \frac{\partial E_{\text{ens}}}{\partial t} = E_{q_2} - E_{q_1} = \mu$$

- A minimum of straight lines is concave.
- $\Delta E \geq 0$: the orbital-relaxation penalty.
- μ drifts slowly down along the branch.

Branch average: $\int_0^1 \mu\, dt = E_{q_2}^{\text{UHF}} - E_{q_1}^{\text{UHF}}$.

The Hartree–Fock analogue of Janak's theorem: J. F. Janak, *Phys. Rev. B* **18**, 7165 (1978).

Backup: GCUHF, numbers for the potassium atom



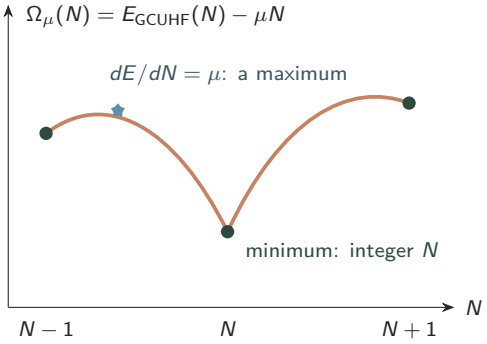
HF/def2-SVP: drift of 0.012 eV

(eV)	IP	EA	Finite diff.	μ, branch
K, HF, SVP	3.980	−0.122	−1.929	−3.980
K, HF, QZVP	4.008	−0.189	−1.909	−4.008
K, MP2, QZVP	4.282	0.262	−2.272	−4.273
K, experiment	4.341	0.501	−2.421	−4.341
O_2, HF	13.28	−1.03	−6.13	−12.5 to −14.2
O_2, experiment	12.071	0.451	−6.26	−12.07

Check: $[E(18.5) - E(18.3)]/0.2 = -3.9786$ eV.
GCUHF at $N_{\mathrm{tgt}} = 18.4$: $\mu = -3.9786$ eV.

Ionization branch, $N - 1 < N_{\mathrm{tgt}} < N$; vertical values; def2 basis sets for K, aug-cc-pVTZ for O_2. Finite differences: $-(\mathrm{IP} + \mathrm{EA})/2$. Experimental branch value: $-\mathrm{IP}$.
G. Fabbro, I.-M. Høyvik (in preparation); NIST ASD; K. T. Andersson *et al.*, *PRA* **62**, 022503 (2000); J. A. R. Samson, J. L. Gardner, *Can. J. Phys.* **53**, 1948 (1975); M. J. Travers *et al.*, *CPL* **164**, 449 (1989).

Backup: what if we fix μ instead of N?



- E_{GCUHF} is concave between integers.
- So Ω_μ is concave there too.
- Its minimum is at an integer end point.
- A fractional solution is a maximum.

At fixed μ, the charge is an integer.
A fractional charge needs a target N_{tgt},
or an explicit, responsive environment.

The Legendre transform between fixed N and fixed μ is not invertible inside an interval. The same relaxation to integer N is reported for particle-breaking HF: R. Matveeva *et al.*, *J. Phys. Chem. A* **127**, 1329 (2023).